

APPENDIX

# Drawings

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## B.1 P&ID

Piping & instrumentation diagram for the containerized BASF pilot water treatment system: drawing no. SWHC-001/00, rev. 0.98, dated 11-11-2022, “BASF WATER Container” (Grant Agreement 969116 — SonixED, H2020-EIC-SMEInst). Shows pretreatment filtration skids (SP-01–SP-4), dosing units for antiscalant/NaOCl/NaOH/HCl/H<sub>2</sub>SO<sub>4</sub> (CIP), tanks, and instrumentation with alarm tags across the treatment train.

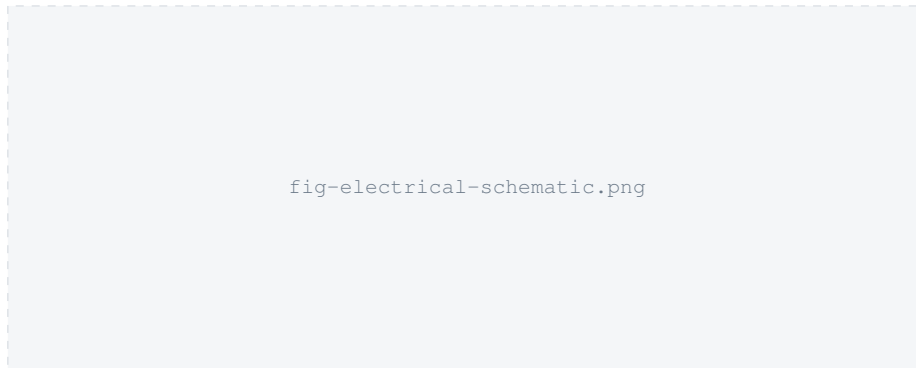


**Figure B.1.** Insert the full P&ID drawing here (source: SWHC-001/00, rev. 0.98) once confirmed representative of the standard SalinBloc™ product, or the product-specific P&ID once available.

HydroVolta's standardization library includes an ANSI/ISA-5.1 P&ID symbol legend, which is the reference standard the symbols on this and other project-specific P&IDs (BASF, Saudi, Australia demo, Morocco) are drawn against.

## B.2 Electrical schematic

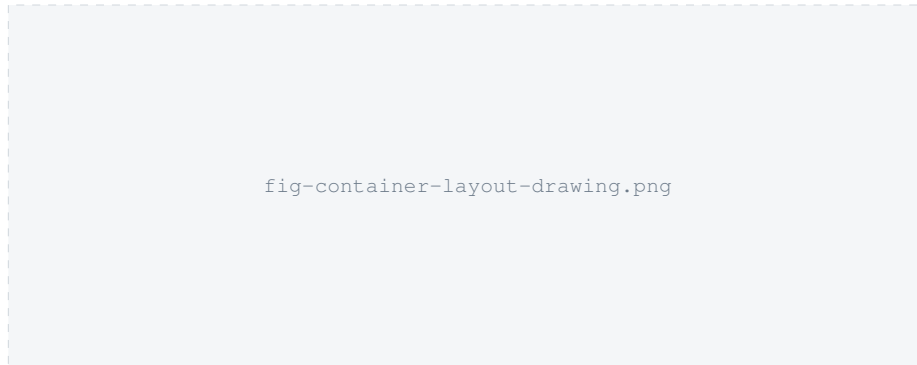
Circuit diagram set “Pretreatment EP” for the pretreatment container, document series C1/I1/I2/E1–E17 (file 31-01-2023 Pretreatment.dsn). Covers the 3-phase incoming supply and breaker/RCD protection (sheet E1), 24 VDC power/control distribution (E2–E3), PLC I/O wiring (E4–E5), terminal wiring (E6–E8), valve/sensor junction boxes (E9–E13), VFD control (E14), sensor/valve wiring standards (E15), dosing pump wiring (E16), and Modbus COM2 wiring (E17).



**Figure B.2.** Insert the relevant electrical schematic sheet(s) here (source: 31-01-2023 Pretreatment.dsn, sheets E1–E17) once confirmed representative of the standard SalinBloc™ product.

### B.3 Container layout

20 ft general-arrangement drawing “2023\_08\_21 SonixED container BASF\_2308CA1” (6058 × 2438 × 2896 mm), and companion skid top-view layout “2023\_08\_22 BASF Skids Top View\_2308CA2” — see chapter 3 for the summary description.



**Figure B.3.** Insert drawings 2308CA1 (general arrangement) and 2308CA2 (skid top view) here, or the standard-product equivalents once available.